

Round 3 Analysis Notebook Export

Public Reaction to a New Educational Technology: Claude for Teachers

Purpose and research question

This notebook documents the reproducible analysis for the Round 3 theme Public Reaction to a New Educational Technology. Claude for Teachers was selected as the case study. The research question is: How did public attention, sentiment, and discussion topics change after the 14 July 2026 launch?

Data sources and cleaning

The raw data was collected from Google News RSS and public Reddit RSS. A GNews API collector was also tested. The raw archive contained 146 records. After restricting the window to 14 July 2026 onward and removing records that were not clearly related to Claude for Teachers and education, 50 records remained.

```
import pandas as pd
raw = pd.read_csv('data/self_collected_round3_raw.csv')
clean = pd.read_csv('data/self_collected_round3_clean.csv')
classified = pd.read_csv('outputs/classified_public_reaction.csv')
print(len(raw), len(clean), len(classified))
```

Output: 146 raw records, 50 cleaned records, and 50 classified records.

Round 2 model reuse

The Round 2 sentiment_model.pkl and topic_model.pkl files were reused without retraining. The inference step used the same normalization approach as Round 2 before calling predict().

```
import html, re, pickle

def clean_text(value):
    text = html.unescape(str(value))
    text = re.sub(r'https?://\S+|www\.\S+', ' URLTOKEN ', text)
    text = re.sub(r'@\w+', ' USERTOKEN ', text)
    return re.sub(r'\s+', ' ', text).strip().lower()

with open('models/sentiment_model.pkl', 'rb') as f:
    sentiment_model = pickle.load(f)
with open('models/topic_model.pkl', 'rb') as f:
    topic_model = pickle.load(f)
model_text = classified['text'].fillna('').map(clean_text)
classified['sentiment_prediction'] = sentiment_model.predict(model_text)
classified['topic_prediction'] = topic_model.predict(model_text)
```

Prediction results

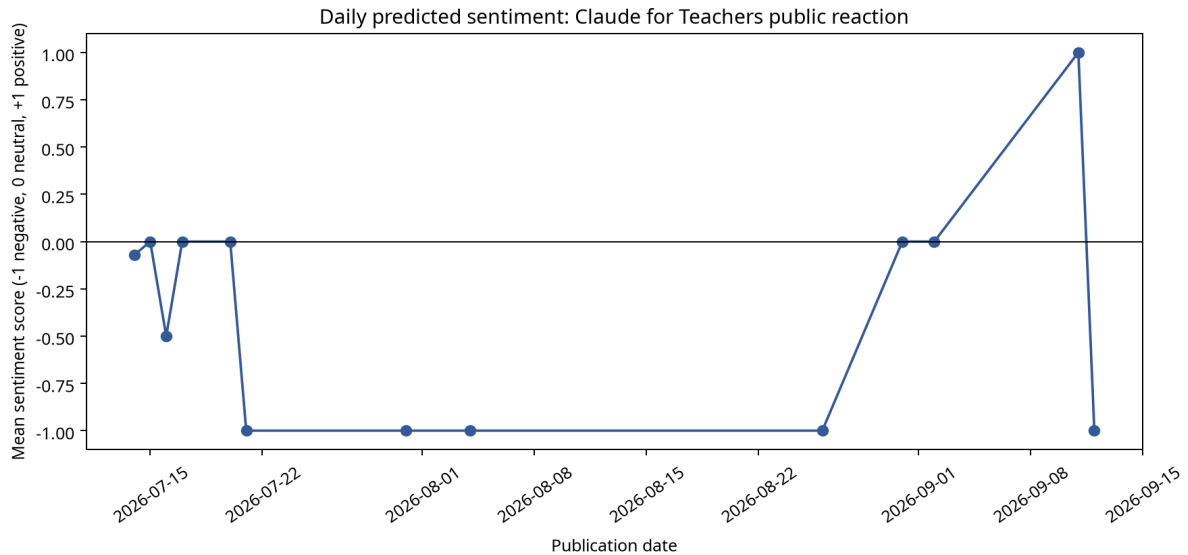
Sentiment	Records	Share
Neutral	39	78.0%
Negative	9	18.0%
Positive	2	4.0%

Topic	Records	Share
Community_Discussion	45	90.0%
Feature_Feedback	4	8.0%
Technical_Issues	1	2.0%

Time-based analysis

```
classified['published_at'] = pd.to_datetime(classified['published_at'], utc=True, errors='coerce')
classified['date'] = classified['published_at'].dt.date
classified['sentiment_score'] = classified['sentiment_prediction'].map({'Negative': -1, 'Neutral': 0, 'Positive': 1})
daily = classified.groupby('date').agg(records=('record_id', 'count'), mean_sentiment=('sentiment_score', 'mean')).reset_index()
```

The analysis identified a move toward negative predictions around 21 July. A later isolated negative prediction occurred on 12 September after a positive prediction on 11 September. These dates contain few records, so they are treated as signals rather than population-wide proof.

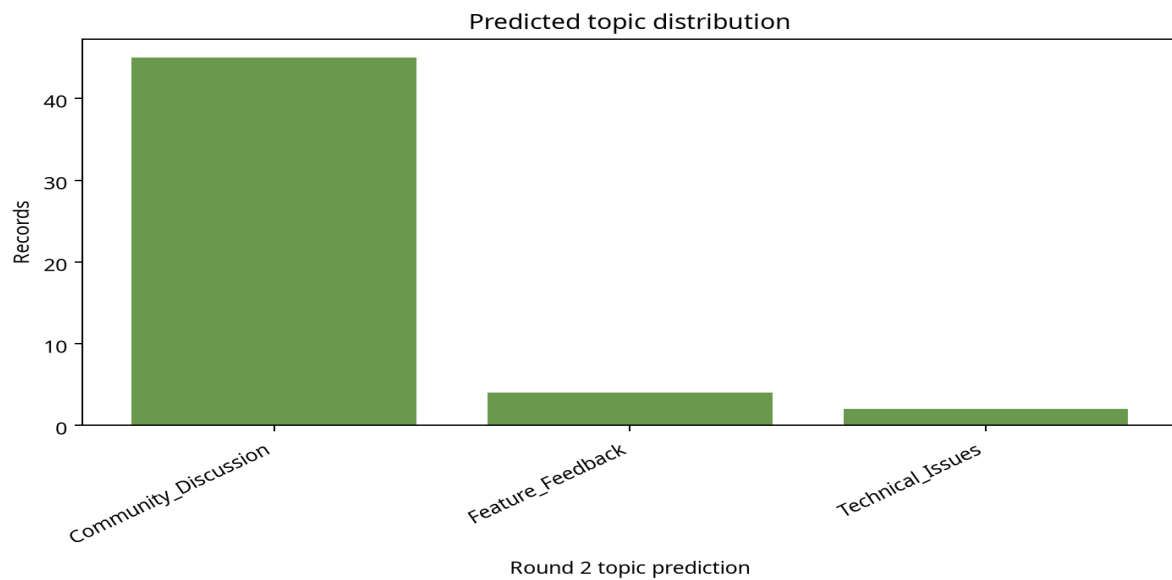
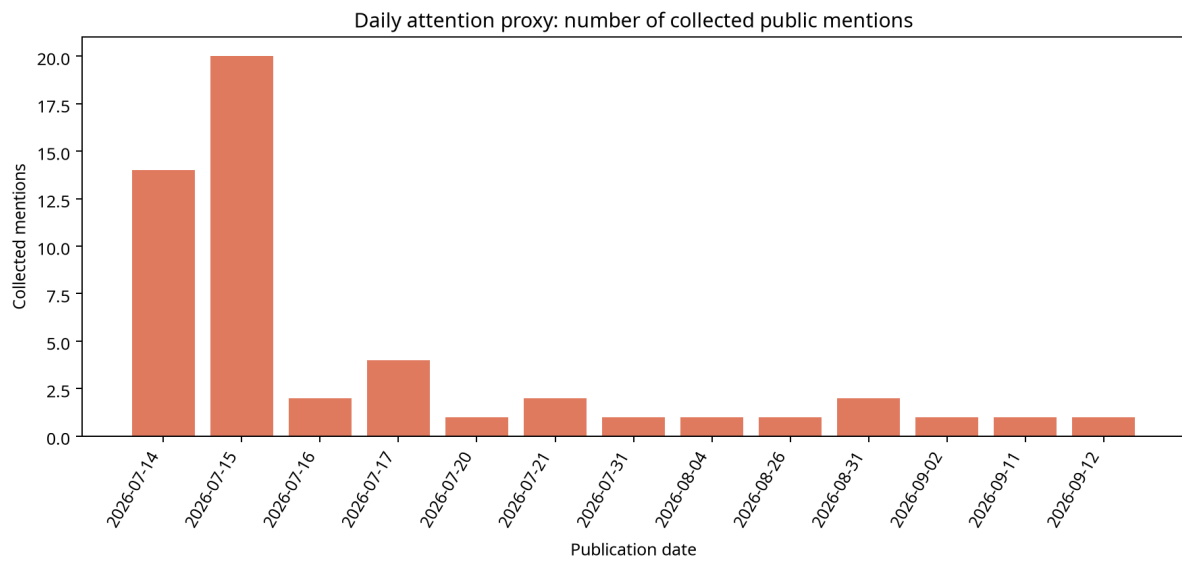


Attention spike analysis

```
classified['attention_proxy'] = 1
daily_attention = classified.groupby('date').size()
print(daily_attention.sort_values(ascending=False).head())
```

The largest attention proxy values were 19 mentions on 15 July and 14 mentions on 14 July. Because public RSS does not consistently expose likes, comments, or shares, this is reported as mention volume rather than true social-media engagement.

Attention and topic visualizations



Interpretation and files used

Public attention peaked immediately after launch. The Round 2 model classified most records as neutral and assigned 90% of the records to Community_Discussion. The notebook uses the cleaned dataset, the two Round 2 model files, the classified output, and the time-analysis CSV files.